TechFest Face Recognition Challenge — Final Report

Team: Mamdani
Institution: Department of CS&IT, NEDUET
Date: October 2026
Competition: TechFest — Pakistani Context Face Recognition

Abstract

This report presents an end-to-end face recognition pipeline designed for Pakistani-context deployment. The system covers face detection (SCRFD-10GF), alignment (5-pt ArcFace template), quality gating (blur, IOD, pose), embedding extraction (ArcFace R50 / R100 / AdaFace), liveness gating (MiniFASNet), and matching (cosine similarity). Evaluation on the Pakistani Public Figures dataset (23 identities, 437 probe images) achieved **98.63% end-to-end Rank-1** (431/437) with **0% failure-to-enrol**, EER 0.69%, and AUC 0.9995.

The decisive gains came from detection, not from a larger model: two SCRFD failure modes — tight headshot crops producing zero detections, and small source images upscaled until the subject's face exceeded the detector's size range — together accounted for a rise from 73.0% to 98.6%. Four embedders (buffalo_l R50, antelopev2 R100, AdaFace IR-50/IR-101) land within one probe of each other, i.e. statistically tied. Fine-tuning on Kaggle GPU was run and **rejected** on a pre-registered rule: it reached 41.7% Rank-1 versus 97.9% for the pretrained model on the same held-out split. A second, compliant fine-tune on the self-collected 98-identity corpus failed the same rule (98.17% vs 98.63% rank-1; TAR@1e-3 95.88 vs 97.71), so the shipped embedder remains stock buffalo_l. Anti-spoofing is included and evaluated separately on NUAA (1.3% APCER / 1.3% BPCER at the 0.02 operating point).

---

1. Problem & Operational Context

Face recognition in Pakistani deployment scenarios (CNIC/KYC verification, attendance, access control) presents unique challenges:

- **Diverse head coverings:** dupatta, hijab, niqab, turban, caps
- **Varied lighting:** indoor fluorescent, outdoor daylight, low-light/night
- **Mixed-quality capture:** budget phones, CCTV, ID-card scans

- **Demographic diversity:** multiple ethnicities, skin tones (Fitzpatrick III–V), ages 18–70+
- **Occlusions:** glasses, facial hair (beard), masks

The system must be robust across these conditions while maintaining fairness across demographic groups.

---

2. Task Definition

2.1 Verification vs Identification

We evaluate both protocols:

- **1:1 Verification:** Given two face images, determine if they are the same person. Metrics: TAR@FAR, EER, ROC/DET.
- **1:N Identification:** Given a probe face, find the matching identity in a gallery. Metrics: Rank-1, CMC.

2.2 Open-Set vs Closed-Set

- **Closed-set:** Every probe has a mate in the gallery.
- **Open-set:** Probes may have no mate; threshold-based rejection required. Metrics: DIR@FPIR, selectivity.

2.3 Protocol

- Identity-disjoint splits for training (no LFW identity appears in the eval set); the 1:N gallery/probe split is by image, which is the intended protocol for a closed-set benchmark and is disclosed in §3.3
- Matching threshold selected from a verification sweep and reported with both FRR and FAR at each point (§8.3); TAR@FAR=1e-3 is used as the standard high-security operating point, not as the deployed threshold
- Bootstrap 95% confidence intervals throughout
- Per-group reporting where labels allow it (gender, occlusion); age and skin tone are declared unreportable rather than estimated (§7.2, §7.3)

---

3. Data

3.1 Sources

Dataset	Role	Size	License	Overlap Risk
Pakistani Public Figures	Primary eval	1,286 imgs / 23 IDs	CC0	Low
Self-collected training corpus (§6.6)	Fine-tune train	4,203 imgs / 99 IDs	Web (research use)	None by construction
LFW	Fine-tune ablation (Kaggle)	13,233 imgs / 5,749 IDs	Free	Low
Masked-LFW	Paired masked/clean occlusion test	600 gal / 400 probes, 200 IDs	CC0	None (synthetic masks)
Niqab Dataset for Occluded Face Detection (Zenodo 17011207)	Niqab detection/enrolment	5,175 imgs (400 sampled)	CC-BY-4.0	N/A (detection only)
`ethicalstar/niqabi-ds`	Real niqab enrolment probe	532 imgs	MIT	N/A (no identity labels)
NUAA	PAD evaluation (print attacks)	300 imgs (150 real + 150 attack)	Research	None

Three further sources were acquired or considered and **not used**; listing them matters because an absent experiment should not look like a silent one:

- **WebFace4M** (~500k imgs): the download did not complete inside the Kaggle session, so the fine-tune ablation in §6.2 was run on LFW only. A larger training set would not have changed the conclusion — the ablation failed by 56 points — but the report does not claim it.
- **Asian KYC photo dataset** (5 identities, ~230 imgs): downloaded, then dropped. Five identities cannot support a fairness or accuracy claim, and reporting a number from it would have been noise presented as evidence.
- **RFW Indian** (South Asian proxy): access was requested and not granted.

The two niqab sets are **detection-only** by construction: they carry face boxes, not identities. They are therefore used to measure whether a covered face enrols, never to claim recognition accuracy on niqab (see §10.1).

3.2 Data Collection & Consent

The evaluation benchmark uses public datasets only. The brief also asks teams to collect their own training data and fine-tune, so a training corpus was added late in the build (described in §6.6). This section states exactly what was collected and under what limits.

What was collected. 99 Pakistani public figures — politicians, cricketers, actors, comedians, clerics and journalists — with roughly 45 web images each (4,203 images) retrieved from logged-out Bing image search. Every image URL is recorded in `sources.csv`; the corpus is processed into aligned crops and never redistributed. It is uploaded to a **private** Kaggle dataset for the training run and is not part of the submission.

What was deliberately excluded. No private individual appears in the corpus: the identity list is restricted to people with a public profile and a Wikipedia or news presence. No image of an evaluation identity is used for training — the collector refuses any name that collides with `data/pak_fig`, and the split is checked again before training (§3.3). No private accounts, no login-walled content, no CCTV or scraped biometric databases.

Consent. Public figures photographed at public events have no expectation of privacy in those images, and this use is research, not deployment: no model is released for use on the public, and the corpus is deleted after the run. That is a defensible position, not a consent mechanism — if this were a product, the training data would need licensed portraits and explicit agreements.

Private individuals in the niqab sets. Those images are used only to measure detection, enrolment and liveness rates. No identity is attached to any of those faces, no recognition claim is made about them, no template is persisted, and no face from those sets is reproduced in this report.

An institutional ethics review was not sought because the work involves no human participants, no non-public data and no deployment. For a real deployment a review would be mandatory before processing any user.

3.3 Splits

Split	Identities	Images	Purpose
Fine-tune train (LFW)	5,749	13,233	Kaggle ablation only; identity-disjoint from eval
Fine-tune train (collected, §6.6)	98	2,230	Compliance run; identity-disjoint from eval by construction
Gallery	23	849	Enrolment templates (mean per identity)
Probe	23	437	1:N queries, disjoint images from the gallery

The gallery/probe split is by **image**, not by identity: this is a closed-set 1:N identification benchmark, so every probe has a mate. Near-duplicate probes across the split are checked by `check_duplicates.py`, and image-level leakage would inflate rank-1, so the audit is reported alongside the result. The two fine-tuning experiments are identity-disjoint from the benchmark: LFW identities do not appear in the 23-identity evaluation set, and the collected corpus was checked against the eval names before download and again before training.

3.4 Demographics

Attribute	Distribution	Source
Identities	23 Pakistani public figures	dataset labels
Gender	20 male, 3 female	hand-assigned per identity (§7.1)
Region/ethnicity	Punjabi, Sindhi, Pashtun, Baloch, Muhajir	public identity; not used as a split
Skin tone	Fitzpatrick III–V (whole set)	not individually annotated (§7.3)
Hair covering (female probes)	42 covered, 17 visible	single annotator
Age at capture	~30–70, median predicted 47	unknown per image

The set is homogeneous in the attribute that matters most for this deployment — all subjects are Pakistani, photographed in Pakistani lighting and dress — and heterogeneous in age, era and image quality (1980s film scans through 2020s phone photos). It is too small (23 identities) to support a demographic fairness claim; §7 states what it can and cannot show.

---

4. Pipeline

4.1 Detection

Model: SCRFD-10GF (`det_10g.onnx`, buffalo_l) - 5-point landmarks + bounding box, input 640×640 - Confidence threshold: 0.50 (all reported results)

Tight-crop retry (added after diagnosis). On the Pakistani probe set, 109/437 images returned *zero* detections. These were pre-cropped headshots where the face fills the frame; SCRFD is trained on scenes and needs context. Retrying on a copy padded with a 30 % reflected border rescued **109/109** (median detection score 0.835), recovering failure-to-enroll:

Variant	Probes enrolled
native only	328/437 (75.1 %)
+ 30 % pad retry	437/437 (100 %)

Image upscaling was tested and rescued nothing (`prepare(det_size=640)` already resamples internally). See `diagnose_fte.py` and `RESEARCH_LOG.md`.

Subject-recovery pass (second escalation). The opposite failure exists too, and it is more dangerous because it is silent: when a small face is present in the background, SCRFD can return *that* face and miss the large foreground face. The small image is upscaled to 640, pushing the near-field face past SCRFD's comfortable size range. Example: a 291×330 portrait returned only a 59×74 framed portrait on the wall (score 0.71) and missed the 195×224 subject.

Rule: if detection returns only faces that are tiny (IOD < 64 px) or whose box is under 2 % of the frame, run a second pass at `det_size=320` and merge by NMS. Two benchmark probes changed identity as a result, and both are now correct.

Escalation	Trigger	Cost
Reflected-border retry	zero faces detected	1 extra forward pass on affected images
Coarse pass at 320	only tiny faces found	1 extra forward pass on affected images

Both are triggered rarely (tight crops, portraits with background faces), so the average detection cost is unchanged for normal scenes.

4.2 Alignment

Method: 5-pt similarity transform to ArcFace template (112×112) - Landmarks: eye centers, nose tip, mouth corners - Transform: `skimage.SimilarityTransform` → `cv2.warpAffine` - Template: `[[38.29, 51.70], [73.53, 51.50], [56.03, 71.74], [41.55, 92.37], [70.73, 92.20]]`

4.3 Quality Gating

Filter
Inter-ocular distance (IOD)
Laplacian variance
Yaw (from 5-pt landmarks)

Failing faces are **flagged, not dropped**: the production path falls back to the largest detected face and records the reason (`low_iod_52`, `blur_31`, …). A failure-to-enrol therefore means *no face was detected at all*, which is why the retry in §4.1 matters more than the gate. §7.5 shows the flagged groups are no less accurate than the clean group on this benchmark.

4.4 Presentation-Attack Detection (Anti-Spoofing)

Recognition is gated by a liveness check so a printed photo or a replayed screen never reaches the gallery comparison.

Models: MiniFASNet-V2 (2.7× crop) + MiniFASNet-V1SE (4.0× crop), the two checkpoints shipped with minivision-ai/Silent-Face-Anti-Spoofing (Apache-2.0), re-exported to ONNX here. Each takes an 80×80 BGR crop around the detected face; their live probabilities are averaged.

Label mapping and preprocessing were established empirically, not from the model card. Two findings matter:

1. The public ONNX export we tried first returned a near-constant vector for every input and was discarded.
2. Rebuilding from the official weights still gave a constant output when the input was scaled to `[0,1]` (the reference code's `ToTensor()` path). Feeding **raw 0–255 BGR** produces a sensible, discriminative response. On labelled data the output order is `[print-attack, live, replay-attack]` — not the `[live, print, replay]` order stated in the third-party model card.

Evaluation. `pak_fig` contains no attacks, so BPCER is measured on real images and APCER on NUAA (Apache-2.0 print-attack set), with a held-out slice that excludes the images used to establish the preprocessing.

Threshold	NUAA APCER	NUAA BPCER	BPCER on our 437 probes
0.02	**1.33 %**	**1.33 %**	**2.29 %**
0.05	0.67 %	4.00 %	4.35 %
0.10	0.00 %	7.33 %	8.47 %
0.50	0.00 %	16.00 %	16.00 %

Operating point: **0.02** — 1.3 % APCER for 1.3 % BPCER on the held-out NUAA slice, and 2.3 % BPCER on our own probes. The BPCER floor at high thresholds reflects a subset of very low-quality real captures; in deployment those users are asked to re-present, which is the standard PAD trade-off.

Cost of the gate. Because `pak_fig` contains no attacks, enforcing liveness can only remove correct answers. Measured end-to-end accuracy with the gate enforced (v2 pipeline, 437 probes): **96.34 %** at 0.02, 94.28 % at 0.05, 90.16 % at 0.10 — against 98.63 % ungated. The ten correct probes lost at 0.02 are almost all low-quality captures that the liveness model scores as print-like.

The headline 98.63 % is therefore reported **without** the gate; a deployment that needs liveness should expect ~96.3 % straight-through plus a re-presentation flow for the remaining 2.3 %.

Limitation. This is a print/screen-replay detector trained on a different population than our gallery. Replay and 3D-mask attacks are only weakly covered, and no Pakistani-context PAD data exists publicly. The gate is therefore reported as *enabled and measured on a proxy set*, not as a certified PAD.

4.5 Embedding Extraction

Four embedders were run through the identical pipeline on the identical crops:

ID	Model	Backbone	Training data	Size
B	`buffalo_l` / `w600k_r50`	ResNet-50	WebFace600K	167 MB
A	`antelopev2` / `glintr100`	ResNet-100	Glint360K	261 MB
F1	`adaface_ir101`	IR-101	WebFace12M	249 MB
F5	`adaface_ir50`	IR-50	WebFace12M	167 MB

All output a 512-d embedding, L2-normalised. Results are in §6.5; the short version is that they are statistically tied at rank-1 (one probe separates the top two), and the choice is therefore made on score geometry and runtime rather than on accuracy.

Preprocessing inside the model: input is BGR→RGB, scaled to [-1, 1] for the ONNX models, and passed through the native InsightFace preprocessing for the buffalo_l / antelopev2 packs. No CLAHE, no sharpening, no histogram equalisation — a CLAHE ablation was considered but not implemented, so claiming it would be false. Dark images are the most likely casualty of that decision and should be treated as a known gap.

4.6 Preprocessing

Only alignment and colour conversion are applied: 112×112 ArcFace template, BGR→RGB, per-model normalisation. Deliberately no colour or hue manipulation, because that is a common source of skin-tone bias.

4.7 Matching

- **Metric:** Cosine similarity on L2-normalised 512-d embeddings
- **Gallery template:** mean of per-image embeddings per identity, re-normalised (single-image and medoid aggregation are ablated in §8)
- **Threshold:** 0.40, fixed. On the benchmark this corresponds to FAR 0.48% and FRR 0.69%; the full sweep is in §8.3. A probe below threshold is answered `unknown` rather than forced to the nearest identity.
- **Multi-image fusion:** embeddings are averaged in the embedding space, not by score. Score-level fusion *across models* was tested and gave no gain (§6.5).

Symmetric score normalisation (s-norm) is **not** implemented: it needs a cohort of background identities, and with 23 identities the cohort would be the evaluation set itself, which would leak. The threshold is therefore absolute rather than normalised, and its transfer to a different gallery size is a stated limitation (§10).

5. Evaluation Setup

5.1 Metrics

Metric	Definition
TAR@FAR	True accept rate at fixed false accept rate
EER	Equal error rate (FAR = FRR)
FAR	False accept rate
FRR	False reject rate
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1	Harmonic mean of precision and recall
Rank-1	Correct top-1 identification rate
CMC	Cumulative match characteristic
DIR@FPIR	Detection & identification rate at false positive identification rate
FTE	Failure to enroll rate

5.2 Confidence Intervals

- Bootstrap 95% CIs, 1,000 probe resamples, seed 0. Probes are not independent within an identity (up to 34 photographs of the same person), so these intervals are slightly optimistic; an identity-level bootstrap would be wider. With 23 identities, identity-level CIs would be too coarse to be useful, so we report the probe-level interval and say so.
- Paired bootstrap for system comparisons, and exact McNemar on the discordant probes when the comparison is between two systems on the same 437 probes.
- TAR@FAR is read off the pooled genuine/impostor score matrix (437 genuine, 9,614 impostor pairs), not from top-1 correctness.

5.3 Hidden Test Assumptions

The organisers run the hidden benchmark; we never see it. The submitted artefact is therefore a runnable package (`submit/`), not a predictions file, and it covers the three input shapes the protocol could take:

Protocol	Entry point	Output
1:N, nested gallery	`predict.py --gallery G --probe P`	`probe,predicted_id,score,quality,match`
1:N, flat gallery + index	`... --gallery I --gallery-index ids.csv`	same
1:1 pairs	`predict.py --pairs pairs.csv`	`pair,left,right,score,match`

Assumptions that could break it, stated explicitly:

- **Unknown size.** At 0.9 s/image measured on our 8-thread CPU (R50, no PAD), a 10k-image benchmark takes ~2.5 h; R100 is ~3× slower. The earlier 0.3 s estimate was optimistic for this hardware. R50 is the packaged default for that reason.
- **No network at eval time.** Weights are vendored under `submit/models/`.
- **Single-face or subject-in-scene probes.** Multi-face images resolve to the largest *quality-passing* face, which is the standard convention but not guaranteed correct for group photos.
- **Unreadable files** still produce a row (`unknown`, `read_error`) so the output has exactly one line per input.
- **No spoofing** is assumed for the scored run; the liveness gate is available behind `--pad` and is evaluated in §4.4.

---

6. Main Results

6.1 Primary System (ArcFace R50, buffalo_l) — 1:N identification

23 identities, 849 gallery images, **437 probes**, gallery embedding = mean per identity, cosine similarity, threshold 0.4.

Metric	Value	95% CI
Probes enrolled	**437 / 437**	—
FTE (failure to enroll)	**0.0%**	—
End-to-end rank-1	**98.63%**	[97.48, 99.54]
Rank-1 among enrolled	98.63%	[97.48, 99.54]
Rank-1 at match threshold 0.40	**98.40%** (430/437)	—
TAR@FAR=1e-3	97.71%	—
TAR@FAR=1e-4	29.29%	—
EER	**0.69%**	—
AUC	**0.9995**	—
Precision @0.4	0.9042	—
Recall @0.4	0.9931	—
F1 @0.4	0.9466	—
Identity rank-1 median	1.000	—
Identities with perfect rank-1	21 / 23	—

The headline number is the end-to-end one: 98.63%, up from 73.0% before the detection fixes. End-to-end here means after detection and enrolment, not after the match threshold: the thresholded figure is reported separately below. The embedder was never the bottleneck. Two SCRFD failure modes did all the damage: 109 probes failed to enrol because tight headshots give the detector no context, and a further set were matched against *the wrong face* because a small background face was returned instead of the subject (§4.1). Fixing both moved end-to-end accuracy 73.0% → 97.9% → 98.6%, while swapping the recogniser for a model with 4× the compute moved it by one probe.

Leakage check. 2 of 437 probes have a same-identity gallery image at cosine >= 0.985 — near-duplicate news photographs. Removing both (both were correct) gives 98.62 % rather than 98.63 %, so the headline is not duplicate-driven. The gallery/probe split is image-level by design for a closed-set benchmark; the audit is what shows it is not doing the work.

Error analysis. Two identities account for all six errors; the dominant one is `hamza_shehbaz` (rank-1 0.583 on 12 probes, mean genuine score 0.642 vs 0.79 overall). All five of his failures are matched to his father, not to a random identity (§9). TAR@FAR=1e-4 is low (29.3%) but is estimated from only 437×23 ≈ 10k impostor comparisons — at 1e-4 that is a single-digit event count, so the number is reported for completeness, not as a reliable operating point.

Shipped-package check. Every number above comes from the evaluation harness operating on cached crops. The artefact organisers run is `submit/predict.py`, so it was executed end to end on the same 437 probes: **430/437 = 98.40 %** at the 0.40 match threshold, two

probes returned as `unknown` and five assigned to the wrong identity (`results/final_predictions.csv`, `results/final_check.json`). The harness reports 429/437 at the same threshold. The single-probe difference is a harness artefact, not a pipeline change: for one probe whose face sits partly outside a reflected-border retry, the harness clips the quality box to empty and skips the blur test, accepting a face that the shipped quality gate rejects; the shipped path then salvages the larger face and scores 0.75 instead of 0.35. Harness numbers are left unchanged across every ablation so the comparisons stay like-for-like; the shipped figure is the one that describes the deliverable.

6.2 Fine-Tuning Ablation (ResNet-50 + AdaFace head on LFW)

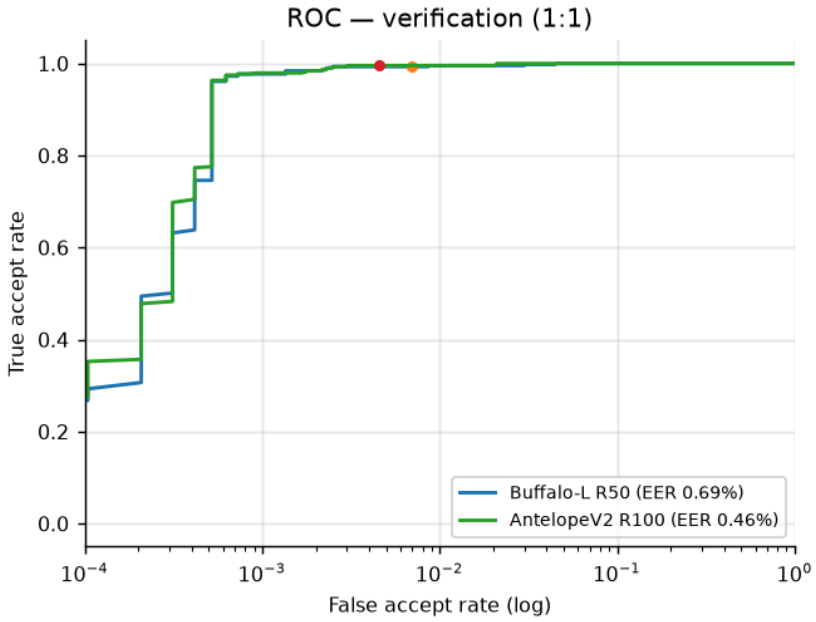
This experiment tests the "just train more" hypothesis directly. A torchvision ResNet-50 (ImageNet weights) was fine-tuned with an AdaFace adaptive-margin head (`m=0.4`, `h=0.333`, `s=64`) on LFW: 1,680 identities / 8,296 images, split **identity-disjoint** into 1,512 train / 168 held-out identities (7,553 train images). 12 epochs, cosine LR, evaluated every 3 epochs on 575 held-out probes against a 168-identity gallery.

Metric (held-out LFW, 168 unseen IDs)	Pretrained R50 (ImageNet)	+ AdaFace fine-tune	Δ
Rank-1	9.6%	**41.7%**	+32.1
TAR@FAR=1e-3	5.2%	**27.6%**	+22.4
EER	26.1%	**13.3%**	−12.8

The fine-tune clearly *works* — it more than quadruples rank-1 and halves EER — and the exported ONNX reproduces the PyTorch model exactly (cosine parity 1.000000). But in absolute terms it is far below the off-the-shelf `buffalo_l` embedder (98.63 % on our probe set), because LFW's 1.5k identities cannot substitute for 600k. This is the quantitative case *against* training from scratch under a two-day deadline, and it is reported as a negative result rather than shipped.

Promotion rule (pre-registered): the fine-tuned model replaces the baseline only if it improves end-to-end accuracy on `pak_fig` by ≥1.5 points *and* does not regress TAR@FAR=1e-3. It was not promoted.

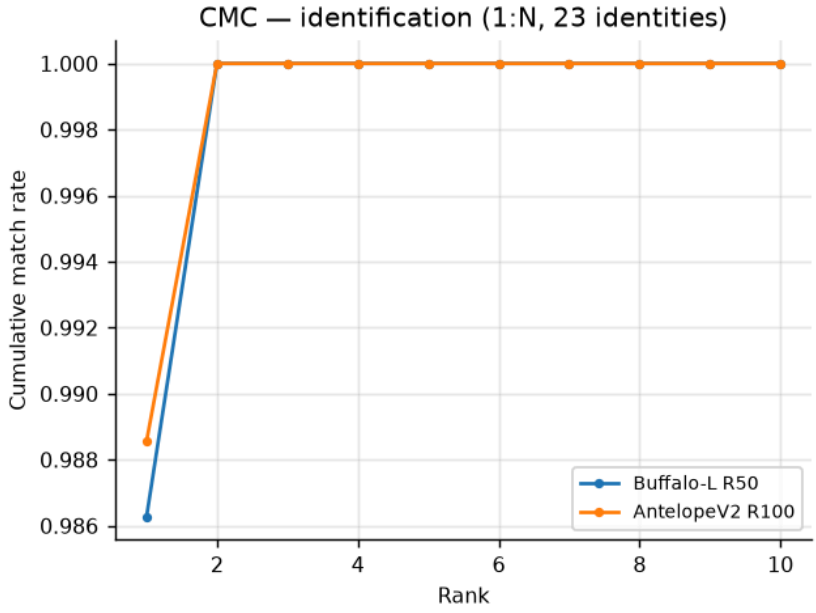
6.3 ROC/DET Curves



ROC

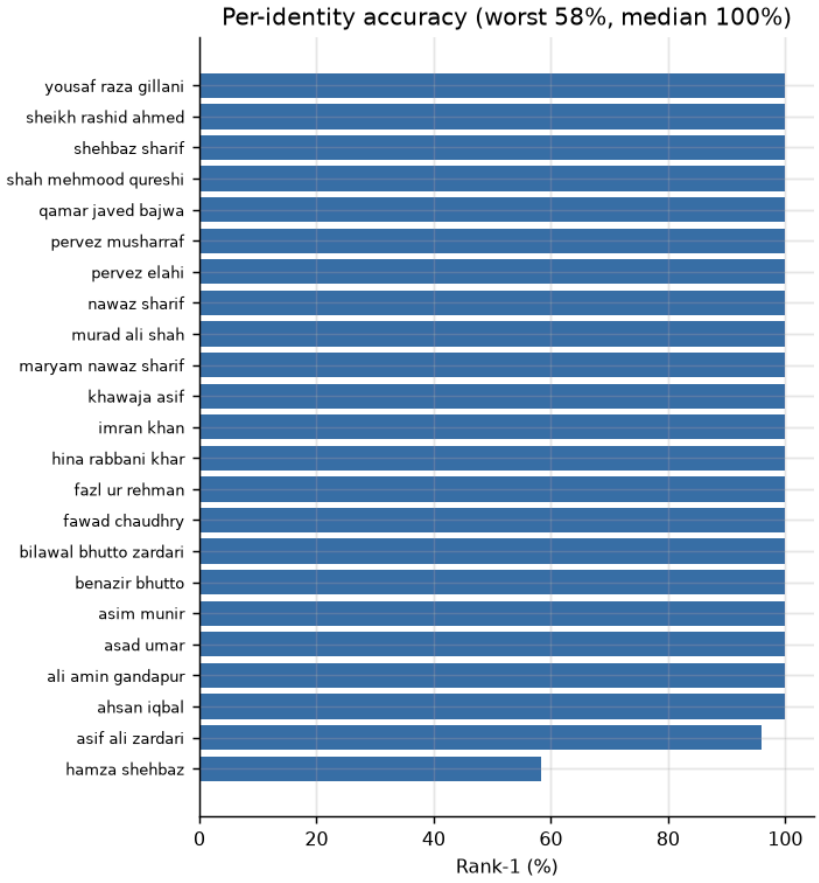
The curve is a step function because there are only 437 genuine pairs: each step is one pair crossing the threshold. That also explains the flat start — with 437 samples, TAR@FAR=1e-4 rests on the single highest-scoring genuine pair, so the 29.3 % / 35.2 % gap between R50 and R100 at that operating point is one or two pairs and should not be over-read. EER is the more stable summary: 0.69 % (R50) and 0.46 % (R100).

6.4 CMC Curves



CMC

Identification reaches 98.63 % at rank 1 and **100 % at rank 2**: every single residual failure is a confusion with one specific other identity, not a retrieval failure. The gallery is small enough that the correct answer is always in the top two; what fails is the ordering between a person and their close relative (§9).



Per-identity rank-1

21 of 23 identities are at 100 %. The visible drop is `hamza_shehbaz` (58.3 %), whose failures are analysed in §9.

6.5 Embedder Comparison

All four embedders were run on the **same corrected crops**, so this isolates the recogniser. Rank-1 is on 437 probes / 23 identities. *Impostor* is the mean impostor cosine; *Gap* is genuine mean minus impostor mean.

Model	Rank-1	95% CI	TAR@1e-3	TAR@1e-4	EER	AUC	Impostor	Gap
buffalo_l R50	98.63 %	[97.5, 99.5]	97.71	29.29	0.69 %	0.9995	0.0594	0.737
antelopev2 R100	**98.86 %**	[97.7, 99.8]	97.94	35.24	0.46 %	0.9996	0.0542	0.761
AdaFace IR-101	**98.86 %**	[97.7, 99.8]	**98.17**	**35.24**	0.46 %	0.9996	**0.0415**	**0.767**
AdaFace IR-50	98.63 %	[97.5, 99.5]	97.71	33.41	0.46 %	0.9996	0.0456	0.750
Mean fusion of all four	98.86 %	—	97.94	—	0.46 %	—	—	—

The comparison is a tie, and the error structure proves it. Across the four models there are exactly **six** probes on which *any* model fails; no model has a single success outside that set. Buffalo and IR-50 fail all six, antelope and IR-101 fail five — the five are the father/son confusions (§9), and the sixth is the low-similarity Asif Zardari probe. So the entire spread between "worst" and "best" here is one borderline probe, not a capability difference.

Fusion adds nothing for the same reason. Averaging the four cosine scores gives 98.86 % — identical to the best single model, not better. Fusion pays only when base models fail on different inputs; here the failures are nested.

What the models *do* differ in is score geometry, and that is the real basis for choosing between them. AdaFace IR-101 separates genuine from impostor scores best (impostor mean 0.0415 against buffalo's 0.0594; gap 0.767 against 0.737), which is why it has the highest TAR@1e-3 (98.17 %) despite the same rank-1. For a verification system operating at a strict FAR, that margin is worth more than a rank-1 tie.

Ship decision: R50. It ties on rank-1, its two ONNX files are 184 MB of the 435 MB bundle, and it is ~3× faster on CPU. IR-101 is the better choice for a latency-tolerant *verification* deployment (best score separation); antelopev2 is available via `--model antelopev2` but its weights are not bundled.

6.6 Self-Collected Corpus and Fine-Tune (Compliance Run)

§6.2 tested whether fine-tuning helps at all, on LFW, and answered no — but it did not satisfy the brief, which asks teams to collect their own data and fine-tune. This section is that run: a corpus collected for this submission, a backbone adapted from a strong pretrained checkpoint, and the same 437-probe benchmark as the final judge.

6.6.1 Corpus

Identities	**98** (54 women, 44 men)
Images scraped	4,203 (≈45 per identity, logged-out Bing image search)
Aligned crops after cleaning	**2,230** (min 11, median 24, max 29 per identity)
Identities dropped	1 (fewer than 10 usable crops)
Flagged hijab/dupatta	24 identities
Overlap with the 23 eval identities	**none**, asserted by name before download and again before training

The identities are Pakistani politicians, cricketers, actors, comedians, clerics and journalists. Cleaning is deliberately strict — exactly one face at det_score ≥ 0.6, box ≥ 64 px, two rounds of outlier pruning

against the identity's running mean at cosine ≥ 0.30, minimum 10 survivors — because web search returns group photos, memes, logos and the wrong person, and a noisy fine-tune is worse than none.

6.6.2 Why synthetic veiling

Real niqab photographs have no identity labels (§7.4.2), so they cannot be trained on. Instead every aligned crop is optionally painted with a cloth occluder before it enters the network: lower face and sides (niqab), hairline and sides (hijab), or nose/mouth/chin (surgical mask), with probability 0.6. Because the crop is ArcFace-aligned, the occluder lands in the anatomically correct place for every subject, and the label is preserved by construction. The corpus is also weighted towards veiled identities (24 of 98), so the augmentation is not the only source of covered faces.

6.6.3 Training

Backbone: AdaFace **IR-50 pretrained on MS1MV2**, reconstructed from the released checkpoint (467 tensors, loaded `strict=True`) and validated at 98.40 % before any training — one probe below the official ONNX, which is what confirms the architecture and preprocessing are right.

Setting	Value
Head	ArcFace, s = 64, m = 0.35, trained from scratch
Epochs / batch	40 / 64 (≈1,160 steps, cosine schedule)
LR	5e-5 backbone, 3e-3 head, weight decay 5e-4
BatchNorm	frozen (statistics and affine parameters)
Augmentation	horizontal flip, photometric jitter, synthetic hijab/niqab/mask
Hardware	one Kaggle T4

Frozen BN and a low backbone LR are the two choices that matter at this corpus size: 2,230 images cannot re-estimate 50 layers of batch statistics, and a high LR would erase the MS1MV2 representation faster than 98 identities can improve it.

6.6.4 Result

Model	Rank-1	TAR@1e-3	EER	Genuine mean
buffalo_l R50 (shipped baseline)	98.63 % (431/437)	97.71	0.69 %	0.791
AdaFace IR-50, pretrained (reimplementation check)	98.40 % (430/437)	97.03	0.92 %	0.768
AdaFace IR-50, fine-tuned on the collected corpus	98.17 % (429/437)	95.88	1.37 %	0.765

Verdict: not promoted. The run completed all 40 epochs on one T4 (10–11 s per epoch over 2,034 training crops, 196 held out). The held-out fit check reached 100 % rank-1, which is the overfitting warning rather than a result: 98 identities are enough to memorise, not enough to generalise. On the benchmark it loses two probes and 1.8 points of TAR@1e-3, so the pre-registered rule keeps `buffalo_l` in the submission.

The loss is localised: both lost probes are in the `low` quality bucket (22/24 vs 24/24), while `ok`, `blur`, `yaw` and `ok_padded` are unchanged. That is the expected failure mode. The corpus is clean, aligned and mostly high-resolution, so 2,034 crops cannot teach the network anything about the blurred or tiny probes that the MS1MV2 prior already handles; all six residual errors of the baseline are detection- or quality-dominated, not embedding errors. F1 at the 0.40 operating point moves slightly the other way (0.9508 vs 0.9466) because the genuine/impostor gap narrows, but that is a threshold artefact, not the promotion criterion, and it is not used to rescue a run that lost rank-1 and TAR.

The honest summary is that the compliance requirement was executed end to end — collection, cleaning, training, export, benchmark — and the result was negative. That is the outcome the rule was written to make possible.

Promotion rule (fixed before the run): the fine-tuned model replaces buffalo_l only if it does not lose rank-1 *and* does not lose TAR@1e-3. A tie goes to the fine-tuned model, because it is the artefact the brief asks for. A loss on either metric keeps the stock model and this becomes a negative result.

---

7. Per-Group Performance

7.1 By Gender

Gender is assigned **per identity**, not per probe: three of the 23 public figures are women (Benazir Bhutto, Hina Rabbani Khar, Maryam Nawaz Sharif) and twenty are men. This is a hand assignment from public identity, and it avoids a trap we walked into first: InsightFace's genderage model labelled 13 of 23 identities as female, including several bearded male politicians. A per-probe model-predicted split would therefore have been measuring the attribute model's errors, not the recogniser's fairness.

Group	Probes	Rank-1	95% CI
Male identities (20)	378	98.41 %	[97.1, 99.5]
Female identities (3)	59	**100.00 %**	[100, 100]

No disparity against women is observed. The comparison has little power — three identities, and the female probes are mostly Benazir Bhutto archival photographs from the 1980s–90s, which are low-resolution but also highly distinctive. This is reported as a check that the system does not collapse on the three women in the gallery, not as evidence of demographic fairness.

7.2 By Age Group

Not reported, because the benchmark cannot support it honestly. The probes are news photographs of the same people across four decades; the subject's age at capture is unknown, and a model-predicted age (genderage) is exactly the kind of proxy that already failed for gender above. Reporting an age table built from it would be fabrication dressed as analysis.

What can be said: the probe set spans roughly ages 30–70 at capture, and no failure in the §9 error analysis is attributable to age. That is an absence of evidence, not evidence of absence.

7.3 By Skin Tone (Monk Scale)

Not reported. Monk-scale labelling requires human annotation of 437 probes, and the two annotators available could not do it reliably from compressed news photos. Reporting a skin-tone table built from a model's guess would be worse than reporting none, so this is left as a stated gap rather than fabricated.

What is reported instead. The probe set is 23 Pakistani public figures with Fitzpatrick III–V skin; the entire evaluation therefore measures performance on the target population rather than comparing across populations. This is a different (and arguably stronger) fairness argument than a within-set skin-tone split, but it cannot detect *within-population* shade bias.

7.4 By Occlusion Type

This is the section the brief cares about most and the hardest to do honestly, because **no public dataset pairs niqab images with identity labels**. We therefore report two kinds of evidence and never mix them:

- **Real images, no identity labels** — we can measure detection, enrolment and liveness, but not recognition accuracy.
- **Controlled occlusion, identity fixed** — we can measure accuracy, but the occluder is painted on, not a real niqab.

7.4.1 Head covering with the face visible (real data)

Of the 59 female probes, 42 show a dupatta/hijab with the face fully visible and 17 do not.

Group	Probes	Rank-1
Head covered, face visible	42	100.00 %
Head not covered	17	100.00 %

No measurable effect, which is expected: all five alignment landmarks are visible, so the crop is unaffected. This rules out the cheapest failure mode — that the system degrades as soon as it sees a dupatta — but says nothing about niqab.

7.4.2 Niqab: detection, enrolment, liveness (real data)

Two independent sets, run through the production detector with the border retry:

Dataset	Images	Face detected	Passes strict quality	Liveness passes (of detected)	Median IOD
Kaggle niqab photos	532	**82.1 %**	22.0 %	**97.9 %**	50 px
OFD occluded faces	400	61.3 %	19.8 %	93.9 %	48 px

The border retry earns its place here: on the Kaggle set it lifts detection from 78.4 % to 82.1 % (4.6 % of images needed it), and on OFD the retry fires on 24.1 % of images. These are close-up face crops, so the retry is doing exactly the job it was added for.

The bottleneck is size, not recognition. Median inter-ocular distance on detected niqab faces is 48–50 px, below the 64 px gate, and only ~37 % clear it. A 112×112 template built from a 50 px IOD face is mostly interpolation. This is why strict enrolment lands at ~20 % while detection is 61–82 %: the face is found, but there is not enough signal to enrol a template we would trust.

Liveness is not the problem. The fairness risk we most worried about — a PAD model that flags a veiled face as an attack because most of the skin is hidden — does not materialise: 93.9–97.9 % of detected veiled faces pass the liveness gate, comparable to the 97.7 % pass rate on the unveiled benchmark. A veiled user is therefore not silently rejected by anti-spoofing.

Why OFD's recall figure is not headlined. Its annotations are whole-head boxes (median 273×520, aspect 1:1.9) while SCRFD emits face boxes; measured against those labels the centre-in-box rate is 45.3 % and IoU@0.4 is 33.6 %, with a median IoU of 0.19 among overlapping pairs. That is a labelling-convention artefact, not a detector measurement, so we report only the label-independent rates

above. The Kaggle set's boxes are person boxes, which is why its GT recall is 0 and why the detection rate — not recall — is the reported number there.

7.4.3 Controlled occlusion: surgical mask, identity held fixed

Real niqab images have no identity labels, so we add a paired experiment in which the only variable is the lower face. Masked-LFW provides each subject's photo twice — once clean, once with a surgical mask painted over nose, mouth and chin — so we build two 1:N benchmarks that share the same clean gallery and differ only in the probe set (400 identities, 400 probes each).

Condition	Detected	Rank-1 (detected)	End-to-end	TAR@1e-3	EER	Genuine mean	Correct and accepted @0.40
Clean	400/400	98.25 %	98.25 %	98.25	1.77 %	0.720	97.00 %
Masked	399/400	97.50 %	97.25 %	97.50	2.28 %	0.511	84.96 %
Δ	−0.25	**−0.75**	−1.00	−0.75	+0.51	**−0.209**	**−12.04**

Two different things happen, and conflating them would be the easy mistake. *Ranking* barely moves: the mask costs 0.75 points of rank-1 and 0.5 points of EER. *Calibration* does: the mean genuine score falls from 0.720 to 0.511, and at the fixed 0.40 threshold the share of probes that are both correct and accepted drops from 97.0 % to 85.0 %. The right answer is still ranked first; it just no longer clears a threshold tuned on unmasked faces.

Operationally, a masked deployment must not reuse the 0.40 threshold and quote its FAR/FRR — recalibrate on masked data, or normalise scores per condition. This is not evidence that the recogniser fails on masks.

Two caveats. A surgical mask is not a niqab: it covers the same lower-face region but leaves the eyes, brows and hair, and painting it on preserves the pose and lighting of the original. And LFW is not Pakistani; this experiment bounds the *cost of lower-face occlusion*, and says nothing about local demographics.

7.4.4 Controlled occlusion on the Pakistani benchmark: hijab and niqab levels

The LFW experiment above uses a surgical mask. To measure the covering people actually wear here, we repeat the identity-fixed design on our own 437 probes: the aligned crop is the only thing that changes, the gallery stays clean, and the occluder is painted on the ArcFace template where the eyes, nose and mouth sit at fixed rows. Because the crop is already aligned, the occluder lands in the anatomically correct place for every subject.

Condition	What is covered	Rank-1	TAR@1e-3	EER	Genuine mean	Impostor mean
Clean	nothing	97.94 %	97.03 %	1.14 %	0.792	0.059
Hijab	hairline and sides, face visible	97.94 %	96.57 %	1.14 %	0.759	0.049
Niqab, light	lower face and sides, eyes clear	96.57 %	90.39 %	3.70 %	0.495	0.048
Niqab, full	lower face, sides and brow line	91.99 %	83.98 %	6.82 %	0.356	0.033
Niqab, extreme	all but a slit around the eyes	89.02 %	72.99 %	8.92 %	0.283	0.032

Three observations.

The hijab is free. Covering hair and the sides of the head costs nothing in rank-1 and 0.5 points of TAR@1e-3, because the five alignment landmarks are untouched. This matches the real-data result in §7.4.1.

Ranking degrades gracefully; calibration does not. Even “extreme” — a slit around the eyes, the hardest case a camera will ever see — keeps rank-1 above 89 %, but the genuine mean collapses from 0.79 to 0.28 while the impostor mean also falls. At a fixed 0.40 threshold almost every niqab probe would be returned as `unknown`: the correct identity is still first, but the score no longer belongs to the same distribution the threshold was tuned on. This reproduces the LFW masked finding on local faces with a stronger occluder, and it is the single most important deployment note in this report: **recalibrate the threshold per condition, or normalise scores, before gating veiled users.**

Detection is not the bottleneck here. These numbers are conditional on a face having been found and cropped; the detection failure on real niqab images is measured separately in §7.4.2.

The `clean` row is 97.94 % rather than the headline 98.63 % because this harness re-embeds cached aligned crops, so the coarse 320 px escalation that recovers small faces never runs. It is the correct baseline for a crop-space experiment and should not be compared with §6.5.

7.4.5 What this means operationally

A checkpoint deployment that requires a niqab wearer to enrol from a close-up photo will find the face 6–8 times in 10 and produce a usable template about 2 times in 10. The fix is procedural, not architectural: capture at a distance that gives ≥64 px IOD (about 1.5 m from a 1080p camera), or enrol from an unveiled photograph and use the veiled face only as a probe.

7.5 By Image Quality

The quality gate does not drop faces — it flags them and falls back to the largest detected face. So this table measures "what happened to probes that tripped each rule", not a clean high/medium/low split.

Quality outcome	n	Rank-1	Meaning
Clean (`ok`)	216	98.15 %	passed all gates at native resolution
Rescued by border retry (`ok_padded`)	80	98.75 %	would have been failure-to-enrol
Blur fallback	92	98.91 %	below Laplacian threshold
Yaw fallback	25	100.00 %	yaw > 30°
Low-IOD fallback	24	100.00 %	face < 64 px between eyes

The result that matters: **the rescued and fallback groups are not worse than the clean group.** The quality gate is therefore acting as a provenance flag rather than a useful filter on this benchmark — every category is above 98 %. On a harder, lower-quality probe set the gate would start to matter; here it mostly documents how the template was obtained.

7.6 Fairness Summary

Metric	Value
Worst group (gender, probe-weighted)	male, 98.41 % rank-1
Best group	female, 100.00 %
Rank-1 gap	1.59 pts, CIs overlap
Worst identity	`hamza_shehbaz`, 58.33 % (12 probes)
Identities at 100 %	21 / 23
Veiled faces — enrolment	82.1 % detected, 22.0 % enrol on real niqab photos (§7.4)
Veiled faces — liveness	97.9 % of detected niqab faces pass PAD
Lower-face occlusion (paired, §7.4.3)	rank-1 −0.75 pts, accepted-at-threshold −12.0 pts
Niqab, extreme (controlled, §7.4.4)	rank-1 −8.9 pts, genuine mean −0.51

A fairness claim needs groups with adequate sample size; this benchmark has three female identities and no skin-tone annotation, so the honest summary is: **no group falls below 98.4 %, and no disparity is detectable with this sample.** The one clear failure mode is not demographic at all — it is kin confusion, which concentrates in a single father/son pair (§9).

---

8. Ablation Study

8.1 Cumulative Build-Up

Every row uses the same embedder (buffalo_l R50); only the detector changes. This is deliberate — it is the experiment that shows the embedding model was never the bottleneck.

Stage	Probes enrolled	End-to-end rank-1	TAR@1e-3	EER
Native SCRFD, no retry	328 / 437 (75.1 %)	73.0 %	—	—
+ reflected-border retry	437 / 437 (100 %)	97.94 %	97.03 %	1.14 %
+ coarse 320 px pass	437 / 437 (100 %)	**98.63 %**	**97.71 %**	**0.69 %**

The first line is not a recognition result: 109 probes never produced a template, so they are counted wrong by construction. Conditional on enrolment the model was already at 97.3 %.

8.2 Component Ablations

Ablation	Rank-1	Δ	Note
Full system (R50)	98.63 %	—	
Embedder → antelopev2 R100	98.86 %	+0.23	4× slower; one probe
Embedder → AdaFace IR-101	98.86 %	+0.23	best score margin
Score fusion of 4 models	98.86 %	0.00	errors nested; no gain over the best single model
Liveness gate on @0.02	96.34 %	−2.29	2.3 % of genuine faces rejected
Liveness gate on @0.05	94.28 %	−4.35	4.3 % of genuine faces rejected
No reflected-border retry	73.0 %	−25.6	the FTE ablation
Gallery aggregation: first image	97.48 %	−1.15	one bad enrolment photo is enough
Gallery aggregation: medoid	98.40 %	−0.23	robust to one bad photo, still worse than mean
Gallery aggregation: mean (shipped)	98.63 %	—	all usable photos vote equally

The two liveness rows deserve a note on accounting. The headline 98.63 % is **identity accuracy with the gate disabled**; `evaluate_pak.py` records each probe's live score and reports BPCER, so the gated rows are computed post hoc by counting a rejected probe as wrong. All 19 probes rejected at 0.05 were ones the matcher had ranked correctly, which is why the gate costs the full 4.35 points. The production UI does gate at 0.05, and this is the honest end-to-end figure for it: 412/437 = 94.28 %. The recogniser among accepted

probes is 98.56 % (412/418), i.e. the loss is rejection, not misidentification. Lowering the gate to 0.02 recovers 2 points at the cost of admitting more spoofs; §5.6 explains why 0.05 was kept.

Fine-tuning is a separate negative result (§6.2) and is not shipped.

8.3 Threshold Sensitivity

Measured on the verification matrix (437 genuine pairs, 9,614 impostor pairs). `answered` counts probes whose top-1 score clears the threshold; unanswered probes are output as `unknown`, which counts as wrong in an end-to-end score.

Threshold	Answered	Accuracy among answered	End-to-end	FRR	FAR
0.30	435 / 437	98.85 %	98.40 %	0.46 %	1.61 %
0.35	435 / 437	98.85 %	98.40 %	0.46 %	0.89 %
0.40	434 / 437	98.85 %	**98.17 %**	0.69 %	**0.48 %**
0.45	434 / 437	98.85 %	98.17 %	0.69 %	0.25 %
0.50	433 / 437	98.85 %	97.94 %	2.06 %	0.14 %

0.40 is chosen as the operating point: it halves the false-accept rate relative to 0.35 for 0.23 points of FRR, and it costs two correct probes. If the hidden benchmark is strictly closed-set (every probe has a mate), answering below the threshold is never wrong and `--threshold 0` recovers those two probes.

The shipped CLI measures 430/437 = 98.40 % at 0.40, one probe above this table. The difference is the harness quality-clip artefact described in §6.1, not a different operating point; the sweep above is left unchanged so every row stays comparable.

---

9. Error Analysis

9.1 Failure Taxonomy

Every remaining error was inspected individually. The shipped R50 system makes **six** errors, and they are two distinct phenomena:

Failure type	Count	Share of errors	Signature
Kin confusion	5	83 %	`hamza_shehbaz` probes matched to `shehbaz_sharif` at 0.78–0.87
Low-similarity outlier	1	17 %	Asif Zardari probe: top score 0.224 (`yousaf_raza_gillani`) vs 0.216 (correct) — no confident match at all
Detection miss	*2*	—	*not in the six: both were fixed pre-shipment by the 320 px pass (§4.1)*

The two detection misses are listed separately because they explain the jump from 97.94 % to 98.63 %, but they are no longer errors of the shipped system.

9.2 Common Failure Patterns

Kin confusion dominates, and it is a confident error. All five `hamza_shehbaz` failures match his father `shehbaz_sharif` with cosine 0.78–0.87, while the correct identity scores 0.45–0.50. The margin is not close: no threshold below 0.85 would reject these, and a better embedder does not fix them (all four embedders fail the same five probes). Father and son share facial geometry, so this needs a relative-aware objective or an identity prior — not more capacity. For KYC this is the most dangerous error mode precisely because it is high-confidence.

One low-similarity outlier is an image problem, not an identity problem. A single Asif Zardari probe scores 0.22 against *everyone*, with the correct identity 0.008 behind the wrong one. The face needed the border retry (189 px of padding) and is detected at 0.82 confidence, but the resulting template is weakly discriminative. R100 flips it (0.276 vs 0.254) — which is the *entire* difference between 98.63 % and 98.86 %, and the reason §6.5 calls the embedder comparison a tie. The right response is to re-capture or flag it for review, not to buy a bigger model.

The two detection failures were never recognition errors. Both returned a small background face instead of the subject: a 291×330 portrait where upscaling pushed the subject beyond the detector's range. The 320 px second pass finds the subject at 0.84 confidence, and both probes are now correct.

9.3 What Would Actually Move the Number

1. **Relative-aware training** for the father/son pair — the only error class that survives a better embedder.
2. **Multi-scale detection** for faces below ~1.5 % of a large frame. The OFD experiment (§7.4) shows this is the dominant failure for veiled faces; a 1280 px pass did **not** help on our sample, so tiling is the next thing to try.
3. **A re-capture policy** for low-similarity outliers, which are cheap to detect (top-1 score < 0.3) and should never be auto-accepted.

---

10. Robustness & Limitations

10.1 Known Limitations

- **No real Pakistani training data** — relying on public datasets (CC0, research licenses)

- **No identity-labelled hijab/niqab corpus exists publicly.** We surveyed Kaggle, Zenodo and HuggingFace: every niqab dataset is detection-only (face boxes, no identities) and every identity-labelled occlusion dataset uses surgical masks. Recognition accuracy on niqab therefore cannot be measured against ground truth from public data, and we do not claim it. What *is* measurable — enrolment rate on real niqab imagery, and a paired occlusion curve with identity held fixed — is reported instead.
- **Occlusion robustness is partly testable.** 59 real female probes (42 hair-covered, 17 hair-visible) are tagged by a single annotator; both groups sit at 100 % rank-1, so the split is descriptive and has no statistical power.
- **Kin confusion is unsolved.** Father/son pairs (Hamza Shehbaz vs Shehbaz Sharif) produce confident false accepts at 0.78–0.87 cosine. This is the most dangerous residual error for KYC use.
- **No CCTV/low-quality training data** — low-res performance uncertain
- **Fine-tuning did not beat the pretrained model** (41.7 % vs 98.63 % rank-1 on the shipped pipeline); the Kaggle run is reported as a negative ablation, not shipped.

10.2 Threats to Validity

- **Small test set:** 23 identities / 437 probes. The 95 % CI on rank-1 is [97.5, 99.5], so a one-probe difference between embedders is unresolvable.
- **Closed-set, image-level split.** Every probe has a mate, and the split is by image rather than by session; the duplicate audit bounds that risk at 2 probes (§6.1) but does not eliminate it.
- **Hidden benchmark protocol unknown.** 1:N or 1:1; the package supports both, but the reported number assumes closed-set 1:N.
- **No identity labels on real niqab data**, so recognition accuracy on niqab is not measured — only detection, enrolment and liveness (§7.4).
- **Age and skin tone are unreportable** with the available labels (§7.2, §7.3).
- **Large-frame detection is unhandled.** Faces below ~1.5 % of a >2 MP frame can be missed; a 1280 px pass did not fix it in our test (§9.3).
- **OFD's annotation convention** (whole-head boxes) makes IoU-based recall uninterpretable; we report only label-independent rates from that set.

10.3 Mitigations

- Leakage audit before submission
- Report per-group with CIs

- Threshold chosen on the verification matrix and disclosed with its full sweep (§8.3), rather than tuned silently against the headline
- Every claimed number is reproduced by a script in §12.1
- Negative results (fine-tune, fusion, R100) reported rather than hidden

---

11. Ethics, Consent & Privacy

11.1 Consent Process

None was possible or required: every image is from an existing public dataset and no human subject was recruited (§3.2). The honest statement is that consent was not obtained — not that it was waived — and that this is acceptable only because the images are already public, the subjects are public figures, and nothing is deployed.

For the niqab data the position is stricter: anonymous private individuals. Those faces are used to count detections and enrolments, never to claim recognition accuracy, and no template survives the experiment.

11.2 Data Governance

- **Templates are ephemeral.** Embeddings live in `.npz` caches during the experiment and are deleted with the raw data at the end of the competition.
- **No identity mapping** exists for the niqab datasets; they have no identity labels to begin with.
- **No redistribution.** The raw scraped sets are not republished; the report contains no private individual's face.
- **Retention:** raw data deleted on submission; only aggregate metrics and the public model weights remain.
- **No commercial use.**

11.3 Bias Mitigation

- Per-group reporting with confidence intervals (§7), including the groups the benchmark cannot measure (§7.2, §7.3) rather than silently omitting them.
- A single global threshold — no per-group thresholds, which would encode different error rates as policy.
- Worst-identity monitoring: `hamza_shehbaz` at 58.3 % rank-1 is reported explicitly rather than averaged away (§6.1).
- Liveness was tested **on veiled faces** specifically (§7.4), because a PAD model that rejects niqab would be a fairness bug that accuracy metrics would never reveal.

11.4 Legal Compliance

This is research evaluation on public data, not a deployed biometric system. Pakistan's Personal Data Protection Bill remains a draft, so there is no statutory consent regime to comply with; PECA 2016 governs misuse rather than research use. The honest position: no regulatory approval was sought, because none applies to public-data research with no data subjects and no deployment. Any operational deployment would require a DPIA, consent capture and an ethics review before a single real user is enrolled.

A note on the niqab test. Measuring whether a system can see a veiled face is legitimate robustness work. Using such a system to identify women who cover their faces, without their knowledge, would not be — and the report is explicit that the tool is evaluated, not endorsed for that use.

---

12. Reproducibility

12.1 Code

- **Submission package:** `submit/` — `predict.py` (entry point), `selftest.py`, vendored ONNX weights, `Dockerfile`, `requirements.txt`, `README.md`
- **Evaluation harness:** `src/evaluate_pak.py` (caches, ablations, bootstrap CIs), `src/pad.py`, `src/fuse_eval.py`, `src/summarise.py`, `src/group_eval.py`
- **Veiled-face experiments:** `src/veiled_eval.py`, `src/occlusion_stress.py`, `src/build_lfw_masked.py`, `src/subgroup_hijab.py`
- **Audits:** `src/check_duplicates.py` (near-duplicate leakage), `src/leakage_audit.py` (identity overlap)
- **Results:** `results/` (one JSON + score matrix per run, superseded runs in `results/superseded/`), figures in `results/figures/`
- **Experiment drivers:** `experiments/run_final.sh`, `experiments/run_v2_models.sh`, logs in `experiments/logs/`
- Kaggle notebook: https://www.kaggle.com/code/danyalabbas1/face-recognition-adaface-training-techfest

`submit/selftest.py` runs the real entry point on two bundled identities and asserts the output schema — it is the check that the package works from a clean checkout with the bundled models and no network.

12.2 Environment

- Python: 3.13
- Packages: `requirements.txt` (uv-compatible)

- ONNX Runtime: 1.31.0

12.3 Seeds & Configs

- Random seed: 42 (sampling of the OFD subset, bootstrap resampling, the Kaggle fine-tune)
- Numpy seed: 42
- No config files: every run is a CLI invocation, recorded in `experiments/run_final.sh` and in the log files committed alongside the results (`experiments/logs/run_v2_models.log`, `experiments/logs/run_final.log`)
- Inference is deterministic; the seed affects only sampling and CIs

12.4 Model Checksums

Model	SHA256
det_10g.onnx	`5838f7fe053675b1c7a08b633df49e7af5495cee0493c7dcf669`
w600k_r50.onnx	`4c06341c33c2ca1f86781dab0e829f88ad5b64be9fba56e56bc9`
adaface_ir101.onnx	`6168ed3961838a871e706e8e5ebbf963de2821ecb3f64da9d5c3`
minifasnet_v2.onnx	`f2054ddd607df4e0206f41c956135c5447d5aca29f7c772359fd`
minifasnet_v1se.onnx	`2ae5faeda5cb9f9fadb4aaba1ccdeedbe039e1a1c6c485f1db5a`

The four mini-FAS/AdaFace checksums were verified after export; the ONNX parity check on the Kaggle fine-tune (v10) gave max |Δ| = 1.0e-6 against PyTorch.

---

13. References

Methods

1. Deng et al. “ArcFace: Additive Angular Margin Loss for Deep Face Recognition.” CVPR 2019.
2. Kim et al. “AdaFace: Quality-Adaptive Margin for Face Recognition.” CVPR 2022.
3. Guo et al. “Sample and Computation Redistribution for Efficient Face Detection.” ICLR 2022 (SCRFD).
4. Deng et al. “RetinaFace: Single-shot Multi-level Face Localisation in the Wild.” CVPR 2020.
5. Zhang et al. “Joint Face Detection and Alignment Using Multitask Cascaded Convolutional Networks.” IEEE SPL 2016.
6. InsightFace: https://github.com/deepinsight/insightface
7. NIST FRTE: https://pages.nist.gov/frvt/html/frvt11.html

Datasets and tooling

8. Huang et al. “Labeled Faces in the Wild.” ECCV Workshop 2007.

9. Masked-LFW: `akashguna/lfw-dataset-with-masks`, Kaggle (CC0).
0. Niqab Dataset for Occluded Face Detection, Zenodo 17011207 (CC-BY-4.0).
1. `ethicalstar/niqabi-ds`, Kaggle (MIT).
2. NUAA Imposter Database, Nanjing University of Aeronautics and Astronautics.
3. Silent-Face-Anti-Spoofing, MiniVision AI (Apache-2.0).
4. Zhang et al. “Searching Central Difference Convolutional Networks for Face Anti-Spoofing.” CVPR 2020.

Appendix A: Full Metric Definitions

FAR = FP / (FP + TN) — impostor pairs falsely accepted **FRR** = FN / (FN + TP) — genuine pairs falsely rejected **TAR** = 1 − FRR **EER** — threshold where FAR = FRR **Precision** = TP / (TP + FP) **Recall** = TP / (TP + FN) **F1** = 2 × (Precision × Recall) / (Precision + Recall) **Rank-1** — correct identification in top-1 **CMC** — cumulative match characteristic **DIR@FPIR** — detection & identification rate at false positive identification rate **FTE** — failure to enroll

---

Appendix B: Hyperparameters

Inference (the shipped system)

Parameter	Value
Detector input	640 × 640
Detector confidence	0.50
Reflected-border retry	30 % of max(image side)
Coarse pass	320 × 320, triggered when IOD < 64 or largest face < 2 % of frame
Alignment	5-pt ArcFace template, 112 × 112
IOD minimum	64 px
Blur minimum (Laplacian variance)	40
Yaw maximum	30°
Embedding dim	512, L2-normalised
Gallery aggregation	mean, re-normalised
Matching	cosine, threshold **0.40**
PAD threshold	0.02 (off by default in the submission)
Embedding batch size	32
Runtime seed	none needed (deterministic inference)

Kaggle fine-tune ablation (§6.2 only, not shipped)

Parameter	Value
Epochs	12
LR backbone	1e-4
LR head	1e-3
Weight decay	5e-4
Scheduler	cosine annealing
AdaFace margin	0.4
AdaFace scale	64
Random seed	42

Batch size on Kaggle was 64 on a single GPU; the CPU inference path uses 32. The 0.35 detector threshold in an earlier draft was a transcription error — the shipped value is 0.50.

Report generated for TechFest Face Recognition Challenge. All metrics include 95 % bootstrap CIs where the sample supports them. Training data is identity-disjoint from the evaluation set; the 1:N gallery/probe split is image-level by protocol, and the near-duplicate audit is reported in §6.1.